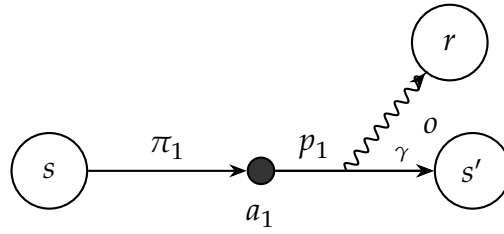


Bellman Equations as Dot Products

(The Pourkavoos-Bellman Equation)



(Above: single step diagram. Next page: the multi-step diagram)

$$g = \vec{\gamma} \cdot \vec{r}$$

$$\vec{o} = \vec{r} + \gamma \vec{v}' \quad \text{recursive}$$

$$q = \vec{p} \cdot \vec{o} \quad = E [g \mid s, a]$$

$$v = \vec{\pi} \cdot \vec{q} \quad = E [g \mid s]$$

$$\vec{\gamma} = \begin{bmatrix} \gamma^0 \\ \gamma^1 \\ \vdots \end{bmatrix}$$

$$\vec{p} = \begin{bmatrix} p_1 \\ p_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} p(s'_1 \mid a, s) \\ p(s'_2 \mid a, s) \\ \vdots \end{bmatrix}$$

$$\vec{r} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} r(s, a, s'_1) \\ r(s, a, s'_2) \\ \vdots \end{bmatrix}$$

$$\vec{q} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} \vec{p}_1 \cdot \vec{o}_1 \\ \vec{p}_2 \cdot \vec{o}_2 \\ \vdots \end{bmatrix}$$

$$\vec{v} = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \end{bmatrix} = \begin{bmatrix} v(s_0) \\ v(s_1) \\ \vdots \end{bmatrix} = \begin{bmatrix} \vec{\pi}_1 \cdot \vec{q}_1 \\ \vec{\pi}_2 \cdot \vec{q}_2 \\ \vdots \end{bmatrix}$$

$$\vec{\pi} = \begin{bmatrix} \pi_1 \\ \pi_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} p(a_1 \mid s) \\ p(a_2 \mid s) \\ \vdots \end{bmatrix}$$

$$\vec{v}' = \begin{bmatrix} v'_0 \\ v'_1 \\ \vdots \end{bmatrix} = \begin{bmatrix} v(s'_0) \\ v(s'_1) \\ \vdots \end{bmatrix}$$

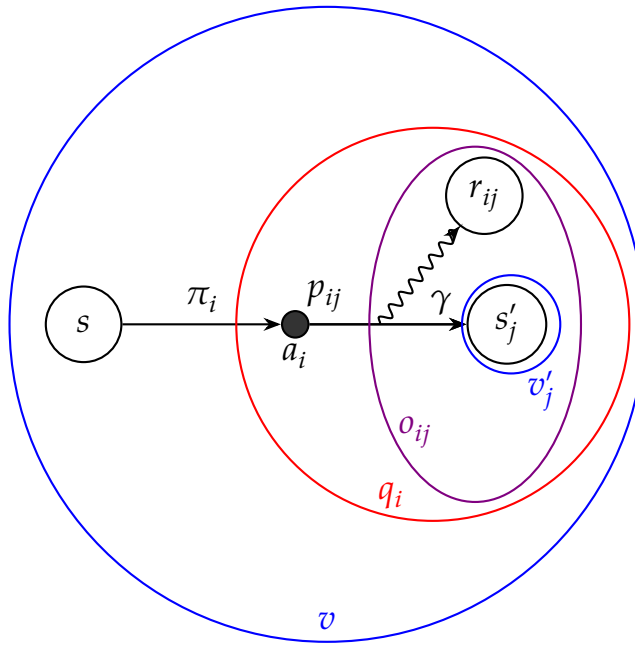
$$\vec{o} = \begin{bmatrix} o_1 \\ o_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} r_1 + \gamma v'_1 \\ r_2 + \gamma v'_2 \\ \vdots \end{bmatrix}$$

Einstein Notation

(implicit summation over repeated indices)

$i = \text{actions}$

$j = \text{next states}$



$$o_{ij} = r_{ij} + \gamma v'_j$$

(elementwise: no summation)

$$q_i = p_{ij} o_{ij}$$

(sum over next states j)

$$v = \pi_i q_i$$

(sum over actions i)

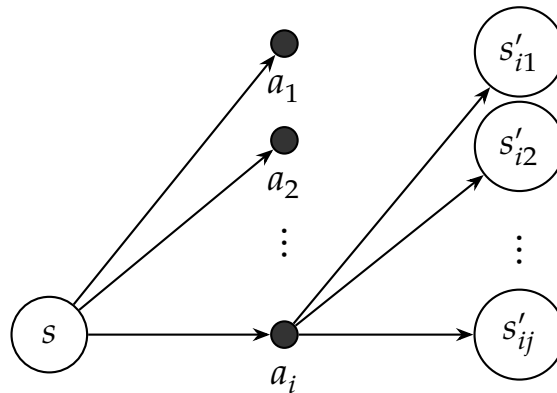
$$v = \pi_i p_{ij} (r_{ij} + \gamma v'_j)$$

(The **Pourkavoos-Bellman** equation: sum over i and j)

Bellman's genius recursive insight:

$$v = E[g \mid s] = E[o \mid s]$$

(The infinite future folds into one step — v' summarizes everything beyond)



The Backup Diagram

GLOSSARY:

r = reward

*(reward comes **after** action a is taken)*

γ = discounting factor

$g = g_0 = r_1 + \gamma r_2 + \gamma^2 r_3 + \dots = \text{return}$ (***single trajectory:** reward + discounted future rewards*)

$o = r + \gamma v' = \text{outcome}$

$s = s_0 = \text{current state}$

$s' = s_1 = \text{next state}$

$v = v(s) = v(s_0)$

$v' = v(s') = v(s_1)$

a = action

p = transition probability, dynamics

v = state value function

q = state-action value function

π = policy